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Measuring Planck's Constant Using LEDs

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Abstract

It seems reasonable to assume that measuring the fundamental constants of the universe requires expensive, high-tech equipment in a professional lab. In reality, some of these constants are readily observable even to an amateur scientist in an undergraduate setting. Planck's constant is particularly interesting as it's present in most of the basic equations of quantum mechanics, despite its physical origin being a mystery at the time of its discovery. In an effort to better understand the physical origin and significance of this constant, I conducted a relatively simple experiment to measure it. Using an understanding of the basic characteristics of light, I observed the behavior of LEDs and approximated a reasonably close measurement of Planck's constant without any expensive equipment. This paper demonstrates how even with amateur experimentation, it is possible to yield accurate approximations of theoretical constants through physical observation.

Introduction

Just as the speed of light defines the scale of general relativity, Planck's constant defines the scale of quantum mechanics. It reveals itself in most of the fundamental quantum equations, indicating its crucial role in describing physics at the subatomic level. Its value was first proposed by German physicist Max Planck in 1900, although he wasn't sure of its physical origin ("Planck's Constant", 2024). He discovered that this particular value (6.626E-34 Js) conveniently solved one of the most fundamental issues in describing energy at the time: the Ultraviolet Catastrophe. At the turn of the 20th century, physicists and chemists alike struggled to explain the unusual behavior of blackbody radiation with classical electromagnetism. Blackbodies–theoretical objects which fully absorb all incident electromagnetic radiation and emit radiation at all frequencies– were simulated using near-ideal conditions to represent their behavior as closely as possible. The classical Rayleigh Jeans Law

$$B_{\lambda}(T) = \frac{2ck_{B}T}{\lambda^{4}} \tag{1}$$

describes the intensity of emitted radiation from a blackbody as a function of one of its particular emitted wavelengths and overall temperature (Peverati, n.d.). It was apparent that as wavelengths tended towards 0, emitted radiation approaches infinity, implying that near infinite energy must be emitted by blackbodies radiating at shorter wavelengths. This physical impossibility led Planck to formulate the following equation to describe the spectral radiation of a blackbody:

$$B_{\lambda}(\lambda, T) = \frac{2hc^{2}}{\lambda^{5}} \frac{1}{e^{\frac{hc}{\lambda k_{B}T}-1}} \tag{2}$$

By incorporating the constant h, shorter wavelengths of emitted radiation no longer required near-infinite energy to be released at these wavelengths as they did in (1). Just 5 years later In

1905, physicist Albert Einstein further associated the constant with the range of energy of electromagnetic radiation itself to explain the photoelectric effect. Classical electromagnetism was based on the assumption that light waves are continuous. It was theorized that electrons must be ejected from a metal after enough energy is accumulated from light waves continuously transferring energy (Chodos, 2005). However, experiments showed that electrons were only dislodged once the light reached a minimum frequency, regardless of the length of exposure. Einstein then concluded that light is not a continuous wave, but instead made up of tiny packets of energy which travel in wavelike patterns. These tiny packets are called quanta, and photons are the quanta of EM radiation (Chodos, 2005). Relating Planck's constant to physical phenomena laid the groundwork for modern quantum mechanics, allowing physicists to manipulate the packet-like nature of electromagnetic radiation for a century to follow.

Using Einstein's theory of quantized light, LEDs of varying wavelengths can be used to approximate Planck's constant. The wavelength of light from an LED determines its color in the visible spectrum. Each color of LED corresponds to a different wavelength and directly related frequency (Davidson, n.d.). Typical LEDs consist of two semiconductor materials separated by a thin gap. One of the semiconducting materials–"P-type"–lacks electrons, and the other–"N-type"–has excess electrons to spare. As soon as a voltage is applied and current begins to flow across an LED, electrons jump from the N-type material to the P-type (Davidson, n.d.). One photon of energy is released for each individual electron that crosses this gap. There exists a minimum voltage required to trigger the release of electrons across the gap, referred to as the "threshold voltage". This minimum voltage varies by frequency, so the threshold voltage can be measured and plotted as a function of varying frequencies of light. As will be explained further in the data and analysis section of this paper, the slope of this function resembles a useful relationship in determining Planck's constant.

Experiment

Materials and Apparatus

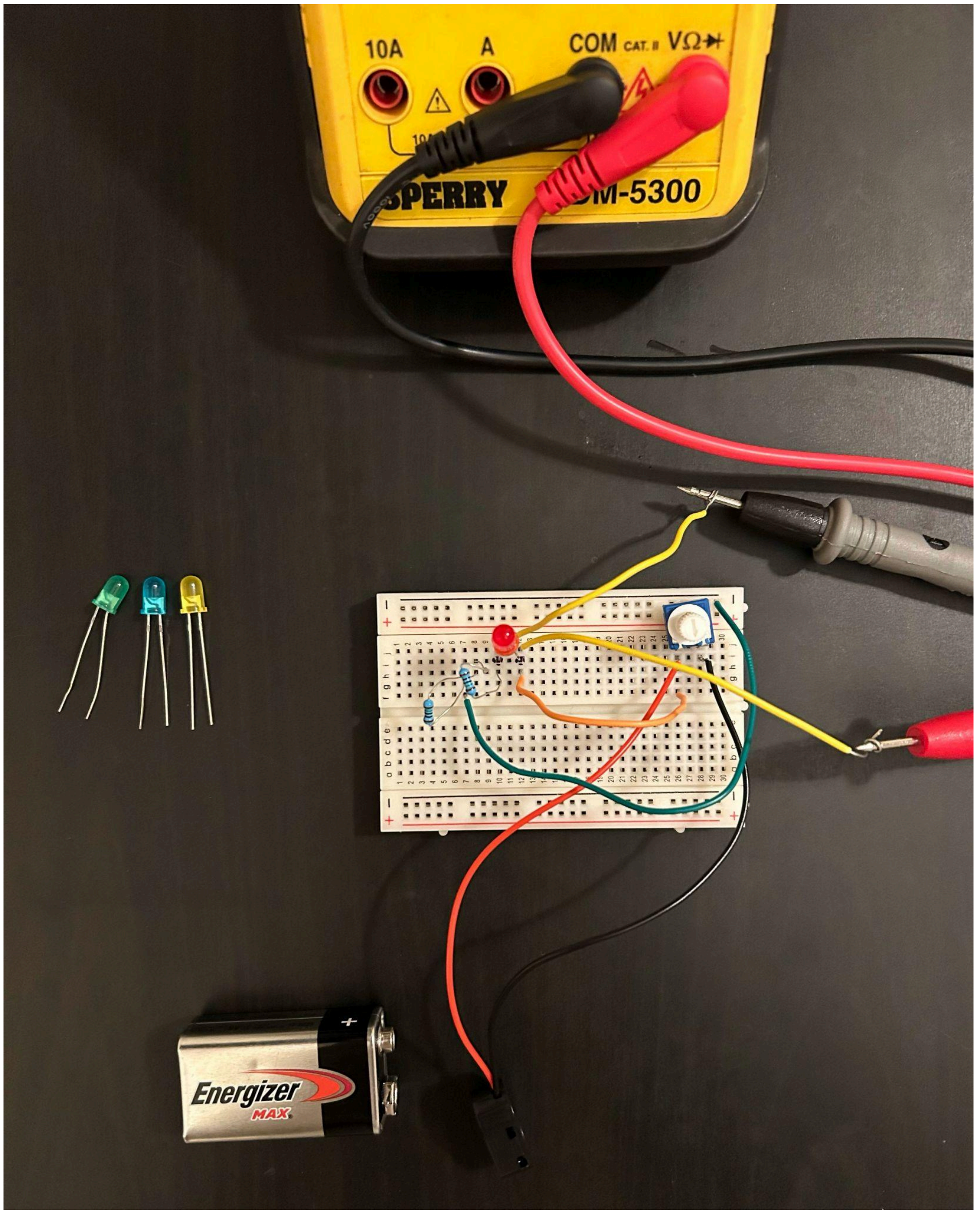
Considering one of the objectives of this experiment is to make use of low-cost and accessible equipment, the necessary components should be easily accessible to most. Four LEDs were used, in four separate colors: red, yellow, green and blue. Any color of LED will be sufficient, since the results depend on the relation between wavelengths as opposed to the kind used. Most manufacturers produce LEDs with the same set of wavelengths for typical colors, so average values of the wavelengths for each color can be used in later analysis. In the case of an unusual LED color or manufacturer, a spectrometer may be used to determine the wavelength, or a diffraction measurement technique if not available. A 10k ohm potentiometer is used to vary the voltage supplied to the LED, and two 150 ohm resistors are used to prevent accidental component damage. Lastly, a multimeter set to measure voltage (a voltmeter will also suffice) is necessary to measure the threshold voltage for each LED. Although not necessary, a breadboard is used to conveniently keep the circuit in place.

Procedure

First, the base circuit was set up, then each color of LED was swapped into the circuit to measure separately. The ground and positive terminals of the multimeter were connected to the circuit via intermediary wires from both terminals of the LED, allowing for threshold voltage readings to be taken more accurately and efficiently.

Figure 1

Circuit setup



Note. Components shown from left to right: two 150 ohm resistors, red LED, intermediary wires for direct connection to the multimeter, and a 10k ohm potentiometer.

After the circuit was set up, the potentiometer was dialed all the way to one side (towards total resistance) to reduce the voltage supplied to the LED to 0 volts. Ensuring the surrounding room was sufficiently dark, the potentiometer was then dialed very slowly until the LED just barely began to glow. To ensure an accurate reading, I set the potentiometer back to full resistance then observed the voltage reading as close to the point of illumination as possible. The voltage supplied to the LED at this point (considered the threshold voltage) was then recorded. This process was then repeated for each LED.

Data and Analysis

The collected data was first organized into a table:

Figure 2

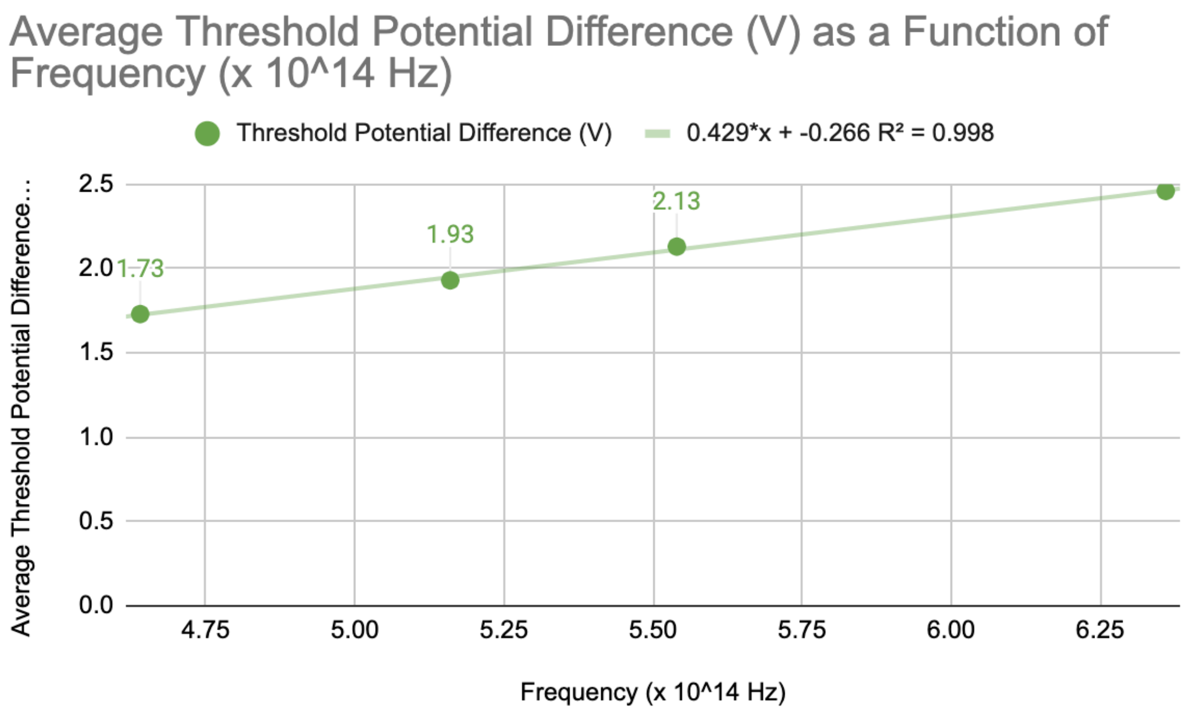
Table of wavelength, frequency, and threshold potential difference for each LED

Color of LED	Wavelength (m)	Frequency (x 10^14 Hz)	Threshold Potential Difference (V)
Red	645	4.64	1.73
Yellow	580	5.16	1.93
Green	540	5.54	2.13
Blue	470	6.36	2.46

After recording the threshold voltage for each LED, the voltages and respective frequencies were plotted.

Figure 3

Graph of threshold voltage vs. frequency



To make use of this data and relate it to Planck's constant, I first considered two equations for the energy of a single photon. First, the energy lost by a single electron as it crosses the gap between semiconductors in the LED (where e is the constant value of elementary charge, and ΔV is the potential difference):

$$E = e\Delta V \tag{1}$$

Second, the generalized energy of a photon based on its frequency:

$$E = hf \tag{2}$$

For each individual electron that passes the gap within the LED, one photon of energy is emitted as E in (1) ("Measuring Planck's Constant", 2008). Consequently, (1) and (2) can be equated to form the following equation:

$$e\Delta V = hf \tag{3}$$

(3) can then be rearranged to solve for Planck's constant:

$$h = \frac{e\Delta V}{f} \tag{4}$$

In this expression (4) for h, elementary charge (e) is a constant, meaning Planck's constant is dependent on the relation $\frac{\Delta V}{f}$. As such, the slope of threshold potential difference as a function of frequency for each LED represents this relationship. The slope of the graph multiplied by the elementary charge should give a reasonable approximation of Planck's constant. Based on a linear form fit to my data, I determined a slope of 0.429. Multiplying this value by the elementary charge gives:

$$h = e \times \frac{\Delta v}{f} = (1.6022E - 19\,C) \times (4.29E - 15\ Js/C) = 6.873E - 34\,Js$$

Comparing this value with the universally accepted value for Planck's constant, the percent error yields:

$$\frac{(6.873E{-}34\,Js) - (6.626E{-}34\,Js)}{6.626E{-}34\,Js} \times 100 = 3.73$$

Through this experiment I determined a value for Planck's constant within 3.7% error from the universally accepted value.

Conclusions

This experiment supports the quantized (packet-like) behavior of EM radiation by relating the theory to real, physical observations. By using a relatively simple and economical setup, I was able to calculate a satisfyingly close approximation of Planck's constant within 3.7% accuracy. Considering the astonishingly small magnitude of this value, I consider this a satisfying result from an at-home experiment with low-cost equipment. By first deriving the theoretical expression for a constant then taking direct measurements through physical observation, I deeply strengthened my understanding of its origin and relevance in the physical universe. This approach (theory first, then physical observation) has deepened my understanding of both Planck's constant and quantum nature more holistically than any textbook explanation had prior. More specifically, this particular experiment has proven to yield approximations of h that are satisfying enough for any amateur physicist to dive deeper into the origin of such a fundamental constant.

Future Work

To improve the accuracy of future iterations of this experiment, a more delicate approach to measuring the threshold voltage could prove helpful. One possibility is to include a phototransistor or photodiode in the circuit, and to connect the setup to an Arduino board. A small program could be written to output the voltage across the LED as soon as the photodiode or phototransistor receives photons. Additionally, the area surrounding this setup must be kept as dark as possible while the LEDs are being observed. During this experiment, a relatively small fraction of ambient lighting entered the room from underneath two doorways. Although I initially assumed the effect to be negligible, this procedure relies heavily on the accuracy of physical observation. Ensuring a darker room equipped with opaque curtains or viewing the LED through a dark, opaque tube will ensure a more accurate measurement of the moment it illuminates. With these modifications employed, this experiment could yield even closer approximations of Planck’s constant.

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